

An Introduction to the Linux Kernel



This article provides an introduction to the Linux kernel, and demonstrates how to write and compile a module.

Have you ever wondered how a computer manages the most complex tasks with such efficiency and accuracy? The answer is, with the help of the operating system. It is the operating system that uses hardware resources efficiently to perform various tasks and ultimately makes life easier. At a high level, the OS can be divided into two parts—the first being the kernel and other is the utility programs. Various user space processes ask for system resources such as the CPU, storage, memory, network connectivity, etc., and the kernel services these requests. This column will explore loadable kernel modules in GNU/Linux.

The Linux kernel is monolithic, which means that the entire OS runs solely in supervisor mode. Though the kernel is a single process, it consists of various subsystems and each subsystem is responsible for performing certain tasks. Broadly, one kernel performs the following main tasks.

Process management: This subsystem handles the process' 'life-cycle'. It creates and destroys processes,

allowing communication and data sharing between processes through inter-process communication (IPC). Additionally, with the help of the process scheduler, it schedules processes and enables resource sharing.

Memory management: This subsystem handles all memory related requests. Available memory is divided into chunks of a fixed size called 'pages', which are allocated or de-allocated to/from the process, on demand. With the help of the memory management unit (MMU), it maps the process' virtual address space to a physical address space and creates the illusion of a contiguous large address space.

File system: The GNU/Linux system is heavily dependent on the file system. In GNU/Linux, almost everything is a file. This subsystem handles all storage related requirements like the creation and deletion of files, compression and journaling of data, the organisation of data in a hierarchical manner, and so on. The Linux kernel supports all major file systems including MS Windows' NTFS.